



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Set Notation & Venn Diagrams

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Your notes

Set Notation & Venn Diagrams

Set Notation

What is a set?

- A set is a **collection of elements**
 - Elements could be anything
 - Numbers, letters, coordinates, ...
- You could describe a set by writing its elements inside **curly brackets** {}
 - $\{1, 2, 3, 6\}$, is the set of factors of 6
- If the set of elements **follow a rule** then you can write this using a **colon inside the curly brackets** $\{ \dots : \dots \}$
 - The bit before the colon is the type of element
 - The bit after the colon is the rule
 - $\{x \text{ is a positive integer} : x^2 < 30\}$ is the set of positive integers which, when squared, are less than 30
 - This is equal to $\{1, 2, 3, 4, 5\}$
 - The colon is often read as '**such that**'
 - If **no type** is specified, x can take **any value** (fractions, decimals, irrationals, ...)
 - $\{x : x^2 < 30\}$ means any value whose square is less than 30
 - $\{(x, y) : y = mx + c\}$ would mean the coordinates (x, y) where $y = mx + c$
 - I.e. The set of all possible coordinates that lie on the line $y = mx + c$
 - A colon can also be replaced by a **vertical bar**
 - $\{x \mid x^2 < 30\}$

What do I need to know about set notation?

- $\mathcal{E}$ is the **universal set** (the set of **everything**)
 - For example, if we are only interested in factors of 24 then $\mathcal{E} = \{1, 2, 3, 4, 6, 8, 12, 24\}$
 - You may see alternative notations used for $\mathcal{E}$



Your notes

- $\cup$ is a common alternative (different to $\cup$ for union!)
- ξ or the Greek letter ξ (xi) may also be seen
- We use **upper case** letters to represent **sets** ($A, B, C, \dots$) and **lower case** letters to represent **elements** ($a, b, c, \dots$)
- $n(A)$ is the **number of elements** in set A
 - For example, if $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 4, 9\}$, $B = \{1, 2, 3, 4, 5, 6\}$
 $n(A) = 3$, $n(B) = 6$
- $\in$ means "is an element of"
 - And $\notin$ means "is not an element of"
 - E.g. $3 \in \{1, 2, 3\}$ and $4 \notin \{1, 2, 3\}$
- $\emptyset$ is the **empty set**
 - This is the set which does not contain any elements
- $A \subseteq B$ means " A is a **subset** of B "
 - Set A is a subset of set B if all the elements of A are also present in B
 - E.g. If $A = \{1, 2, 3\}$ and $B = \{0, 1, 2, 3, 4\}$, then $A \subseteq B$
 - $A \not\subseteq B$ means " A is not a subset of B "
- $A \cap B$ means the **intersection** of A and B (the **overlap** of A and B)
 - This is the set of elements that are in **both** set A and set B
 - E.g. If $A = \{1, 2, 3, 4, 5\}$ and $B = \{-1, 1, 4, 7, 8\}$, then $A \cap B = \{1, 4\}$
- $A \cup B$ means the **union** of A and B (**everything** in A **or** B **or** both)
 - This is the set of elements that are in **at least one** of the sets
 - This **includes** elements in both sets (in the intersection)
 - E.g. If $A = \{5, 6, 7, 8\}$ and $B = \{3, 7, 11\}$, then $A \cup B = \{3, 5, 6, 7, 8, 11\}$
- A' means the **complement of A**
 - It is the **set** of all elements in the universal set $\mathcal{E}$ that are **not in A**
 - E.g. If $\mathcal{E} = \{1, 2, 3, 4, 5\}$ and $A = \{1, 3\}$, $A' = \{2, 4, 5\}$



Your notes

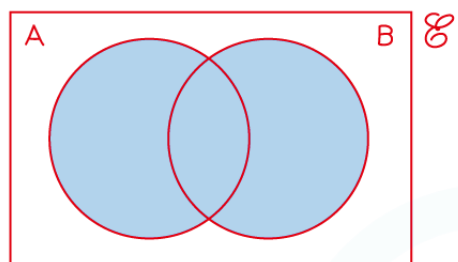
Sets & Venn Diagrams

What is a Venn diagram?

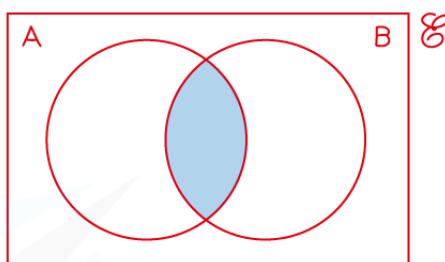
- A Venn diagram is a way to illustrate **all the elements within sets** and any **intersections**
- A Venn diagram consists of
 - a **rectangle** representing the **universal set** ($\mathcal{E}$)
 - a **circle** for each **set**
 - Circles may or may not overlap depending on which **elements** are shared between **sets**

What do the different regions mean on a Venn diagram?

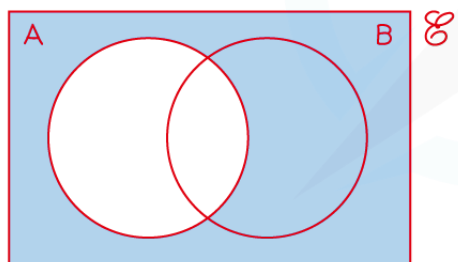
- $A \cap B$ is represented by the region where the A and B circles **overlap**
- $A \cup B$ is represented by the regions that **are in A or B or both**



$A \cup B$ (UNION)
"A OR B OR BOTH"



$A \cap B$ (INTERSECTION)
"A AND B"



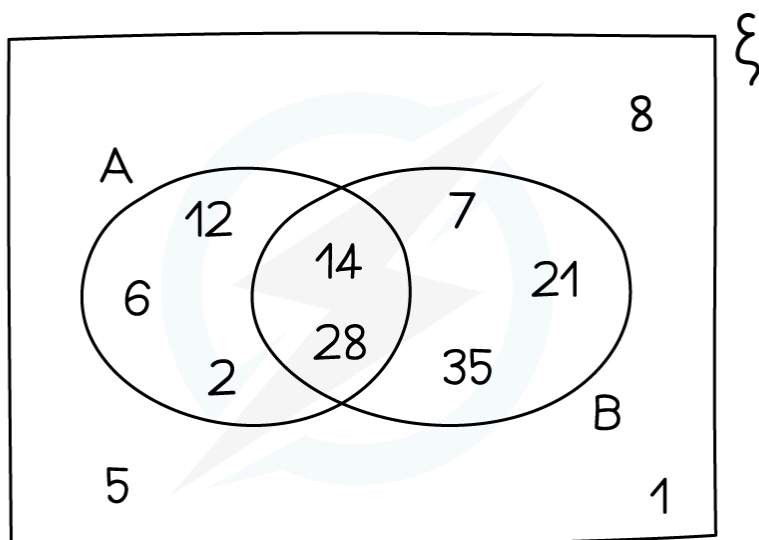
A' (COMPLEMENT)
"NOT A"

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Worked Example

Two sets A and B are shown in the Venn diagram.



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(a) Write down $n(A)$.

The elements of A are anything inside the A circle

$$A = \{2, 6, 12, 14, 28\}$$

$n(A)$ means the number of elements in A

There are 5 elements in A

$$n(A) = 5$$

(b) Use set notation to complete the sentence $\{14, 28\} = \dots$

14 and 28 are the elements that are both in A and B

This means they are in the intersection of A and B

$$\{14, 28\} = A \cap B$$



Your notes



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(c) Write down the elements that are in set $A \cup B$.

$A = \{2, 6, 12, 14, 28\}$ and $B = \{7, 14, 21, 28, 35\}$

$A \cup B$ is the set of elements that are in at least one of the sets

For elements in both, only write them out once

$$A \cup B = \{2, 6, 7, 12, 14, 21, 28, 35\}$$

(d) Jamie states that $A' \cap B' = \emptyset$.

Explain if this statement is correct or not.

The statement means Not A and Not B is equal to an empty set

i.e. There are no elements that are Not A and Not B

However, there are 3 elements that are neither A nor B

There are 3 elements that aren't in A or B

$$A' \cap B' = \{1, 5, 8\}$$

Therefore the statement is incorrect



Worked Example

Consider the following sets.

$$\mathcal{E} = \{\text{letters of the alphabet}\}$$

$$T = \{t, e, a, m\}$$

$$I = \{i\}$$

$$M = \{m, e\}$$

Fill in the blanks with the appropriate set.

(i) ... $\not\subseteq$ T

(ii) ... $\subseteq$...

The symbol $\subseteq$ means "is a subset of", so all of the elements of one feature in the other

The symbol $\not\subseteq$ means "is not a subset of"

The element in I is not in T

$$I \not\subseteq T$$

All the elements of M are also in T

$$M \subseteq T$$



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